DEPARTMENT OF CHEMICAL & BIOMOLECULAR ENGINEERING
NORTH CAROLINA STATE UNIVERSITY

CHE 450 **Homework Set 4**

For Problem 1:

- *Submit written work to Gradescope link similar to previous HW sets*

Problem 1 (8%) Using the heuristics discussed in class, estimate the safe minimum wall thickness of a pressure vessel with a diameter of 2 m which is intended to operate at a pressure of 30 bar. The maximum allowable stress for the material at operating conditions is 160 bar. Assume a weld efficiency of 0.9 and use a corrosion allowance of 0.005 m.

For Problems 2-3:

- *Submit the saved ASPEN Plus files for each problem (file name: ASPEN_HW4_Name_Problem#) to the provided Moodle file upload link*
- *Include your answers as part of a PDF file submission to Gradescope containing:*
    1) *Screen capture of each of the relevant tables/plots as directed*
    2) *Any requested answer(s) to each problem*

Problem 2 (8%) Plot the vapor pressure curves for acetone and methanol up to 200°C on the same plot. Use the NRTL-RK method. Include a screenshot of the resulting plot.

Problem 3 (24%) Plot the Txy diagram for the ethanol-water azeotrope (use mole fraction ethanol for the x-axis) at 1 atm using the **(a, 8%)** ideal, **(b, 8%)** NRTL, and **(c, 8%)** SRK methods. Include a screenshot of each of the resulting plots.

For Problems 4-6:

- *Submit the saved ASPEN Plus files (file name: ASPEN_HW4_Name_Problem#) to the provided Moodle file upload link*
- *Include your answers as part of a PDF file submission to Gradescope containing:*
    1) *Screen capture of 'Input' section of the block and feed stream(s)*
    2) *Screen capture of 'Results' section of each block and outlet stream(s)*
    3) *Final answer(s) to the problem*
    4) *Any additional screen captures requested in the problem statement*

Problem 4 (15%) 100 kg/hr of steam is heated from 350 to 500°C at a constant pressure of 120 bar. How much heat (in kW) is required to be added during this process? Use the STEAM-TA method.

Problem 5 (25%) Pure n-butanol produced at a flow rate of 1000 kg/hr and 2 bar from an upstream distillation process needs to be cooled from 117.7 to 40°C before it can be stored in a tank. Cooling water at 25°C and 2 bar is available to provide cooling, and this cooling water must be returned back to the facility loop at no more than 40°C. Use ASPEN's "Shortcut" heat exchanger model to design a counter-current heat exchanger appropriate for this system using the WILSON method for the n-butanol on the hot side and the STEAM-TA method for water on the cold side. Use a 10°C minimum temperature approach. **(a, 10%)** Design the heat exchanger assuming a cooling water flowrate of 400 kg/hr. Why do you receive an error from ASPEN? Include a screenshot of the appropriate TQ curve in addition to the block and stream information. **(b, 15%)** Now change the cooling water flowrate to 4000 kg/hr and re-run. *(i, 5%)* What is the heat transfer area of the heat exchanger? *(ii, 5%)* What is the heat duty (in kW) for the heat exchanger? *(iii, 5%)* What is the LMTD for the heat exchanger?

Problem 6 (20%) A 1000 kg/hr stream of iso-octane at 100°C and 1 bar enters a counter-current heat exchanger with a heat transfer area of 3 m^2. In the heat exchanger, the iso-octane is cooled by 3500 kg/hr cooling water at 25°C and 2 bar. Use ASPEN's "Shortcut" heat exchanger model in the "simulation" mode using the PENG-ROB method for the iso-octane on the hot side and the STEAM-TA method for cooling water on the cold side. Use a 10°C minimum temperature approach. **(a, 10%)** What is the outlet temperature of the iso-octane (in °C)? **(b, 10%)** What is the outlet temperature of the cooling water (in °C)?